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DEVICE FOR CLEANING AN OPTICAL SURFACE OF AN OPTICAL SENSOR OF A VEHICLE, AND DETECTION SYSTEM

Abstract

A device for cleaning an optical surface of an optical sensor of a vehicle. The device including a first and a second segment respectively having a cleaning fluid circulation channel and a connector, the first and second segments each including a cleaning fluid inlet and at least one fluid diffusion nozzle, the first segment and the second segment being configured to be attached to one another so as to form a circular arc, the connectors having complementary shapes and being configured to interact so as to make it possible to lock the first and second segments relative to one another in an axial direction and in a radial direction.

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Background/Summary

TECHNICAL FIELD

[0001] The invention relates to a device for cleaning an optical surface of an optical sensor of a vehicle, and to a detection system comprising such a cleaning device.

BACKGROUND OF THE INVENTION

[0002] Automotive vehicles are increasingly fitted with optical elements, such as optical position sensors. The function of the optical position sensors is to gather information about the area surrounding the vehicle, in particular to assist the driver in driving and/or maneuvering this vehicle. To this end, an optical sensor is commonly installed on the vehicle so as to collect information about the area surrounding the vehicle. However, such optical sensors are particularly exposed to dirt such as dirty water, dust or other types of spray. This dirt hinders the transmission and reception of information and can disrupt the operation of the optical sensor, or even stop it from operating.

[0003] The use of devices for cleaning an optical surface of optical elements so as to remove this dirt from them has been proposed. These cleaning devices have a complex structure and it may be the case that the cleaning device has a manufacturing defect. The problem is then that the whole of the device is rejected.

[0004] There is a need for a device for wiping an optical surface of an optical element of a vehicle that solves this problem.

SUMMARY OF THE INVENTION

[0005] The aim of the invention is to provide a device for cleaning an optical surface of an optical sensor of a vehicle that stops all of the device being rejected if a manufacturing defect arises.

[0006] For this, the invention provides a device for cleaning an optical surface of an optical sensor of a vehicle, the device comprising a first segment having a first cleaning fluid circulation channel and a first connector, a second segment having a second cleaning fluid circulation channel and a second connector, the first and second segments each comprising a cleaning fluid inlet and at least one nozzle for diffusing the cleaning fluid toward the optical surface, the first segment and the second segment being configured to be attached to one another so as to form a circular arc defining an axial direction passing through the center of the circular arc and a radial direction along a radius of the circular arc, the first connector and the second connector having complementary shapes and being configured to interact so as to make it possible to lock the first and second segments relative to one another in the axial direction and in the radial direction.

[0007] According to a variant, the connectors of segments attached to one another form a detachable snap-fastening means.

[0008] According to a variant, the connectors comprise a projection on one of the segments and a bore in another segment, the engagement of the projection in the bore preventing the segments from moving in translation relative to one another in the radial direction and a tangential direction, which is orthogonal to the radial direction and axial direction.

[0009] According to a variant, the projection can be snap-fastened in the bore, the snap-fastening of the projection in the bore preventing the segments from moving in translation relative to one another in the axial direction.

[0010] According to a variant, the connectors additionally comprise pads and/or lugs on at least one of the segments that interact with the other segment and prevent the segments from rotating relative to one another about the axial direction.
[0011] According to a variant, the connectors comprise tabs on at least one of the segments that engage with the other segment and prevent the segments from moving in translation relative to one another in the axial direction.
[0012] According to a variant, the connectors additionally comprise lugs on at least one of the segments that engage with the other segment and prevent the segments from rotating relative to one another about the axial direction, and/or a stud on at least one of the segments and an orifice in the other segment, the engagement of the stud in the orifice and the snap-fastening of the projection in the bore preventing the segments from rotating relative to one another about the axial direction.
[0013] According to a variant, the connector of one of the segments comprises a tongue which engages in a receiving portion of the other connector of the other segment, the snap-fastening of the tongue in the receiving portion preventing the segments from moving in translation relative to one another in the tangential direction, radial direction and axial direction and about the axial direction.
[0014] According to a variant, the device comprises a passage for an attachment member for attaching the cleaning device to a sensor support, the passage going through the connectors.
[0015] According to a variant, the segments form an annular structure, the structure having two segments attached to one another in pairs, preferably having three segments attached to one another in pairs, preferably having four segments attached to one another in pairs.
[0016] According to a variant, at least some of the segments are identical modules.
[0017] The invention also relates to a detection system comprising an optical sensor of a vehicle and a cleaning device as described above, the device being configured to clean the optical surface of the optical sensor.
[0018] According to a variant, the optical sensor has a cylindrical optical surface, the segments forming an annular structure around at least one part of the circumference of the optical surface.
[0019] All of the preferred embodiments and all of the advantages of the cleaning device according to the invention can be transferred, mutatis mutandis, to the present detection system. The different embodiments can be taken in combination or considered in isolation.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0020] Further features and advantages of the present invention will become apparent from reading the following detailed description, for the understanding of which reference will be made to the appended figures, in which:
[0021] FIG. **1** is a view of a cleaning device;
[0022] FIG. **2** is a view of a segment of the device in FIG. **1**;
[0023] FIG. **3** is a view of an example of connectors;
[0024] FIG. **4** is a view of an example of connectors;
[0025] FIG. **5** is a view of an example of connectors;
[0026] FIG. **6** is a view of an example of connectors;
[0027] FIG. **7** is a view of an example of connectors.
[0028] The drawings in the figures are not to scale. Similar elements are generally denoted by similar references in the figures. In the context of this document, identical or analogous elements may bear the same references. Furthermore, the presence of reference numbers or letters in the drawings cannot be considered limiting, including when these numbers or letters are indicated in the claims.
DETAILED DESCRIPTION OF THE INVENTION

[0029] The invention provides a device for cleaning an optical surface of an optical sensor of a vehicle. The device comprises a first segment having a first cleaning fluid circulation channel and a first connector. The device also comprises a second having a second cleaning fluid circulation channel and a second connector. The first and second segments each comprise a cleaning fluid inlet and at least one nozzle for diffusing the cleaning fluid toward the optical surface. The first segment and the second segment are configured to be attached to one another so as to form a circular arc. In addition, the first connector and the second connector have complementary shapes and are configured to interact so as to make it possible to lock the first and second segments relative to one another in an axial direction and in a radial direction. As a result, the cleaning device is formed of multiple segments attached to one another. This makes it possible to reject one segment independently of another if a defect arises on one of the segments. It is then not necessary to reject all of the device if one segment is faulty.
[0030] FIG. **1** shows a cleaning device **10**. The cleaning device **10** may notably be used in a detection system **11** of a vehicle comprising an optical sensor **12**. The optical sensor **12** makes it possible to collect information on the position of and the area surrounding the automotive vehicle, in particular to assist the driver in driving and/or maneuvering this vehicle. The optical sensor **12** is installed on the vehicle so as to collect information on the area to the front, rear and/or side of the vehicle: the optical sensor **12** is for example installed at the front end and/or at the rear end. The optical sensor **12** is for example a LIDAR, which stands for “light detection and ranging” or “laser imaging, detection, and ranging”.
[0031] The sensor **12** interacts with the surrounding area by means of an optical surface **13**. It may be a protective surface between the optical element and the surrounding area. For example, it may be a surface of a pane fitted between the sensor and the surrounding area, or a surface of a casing which encloses a sensor (such as a face of a LIDAR casing). The surface may be opaque (to the visible wavelengths). The surface may be transparent to the emission and reception wavelengths of the sensor **12**. It is also possible to envisage multiple sensors interacting with the surrounding area by means of a single optical surface.
[0032] The shape of the optical surface **13** can vary according to the location and use of the sensor **12** in the vehicle and depending on how crowded the sensor is. The optical surface **13** may have some portions which are rounded and others which are rectilinear. According to FIG. **1**, the optical surface **13** has a cylindrical shape.
[0033] The device **10** comprises a plurality of segments **14**, specifically at least two segments **141**, **142**. According to FIG. **1**, the device comprises by way of example four segments **141**, **142**, **143**, **144**; it is conceivable that the device **10** comprises a different number of segments. The segments **14** are configured to be attached in pairs. For example, the first segment **141** and the second segment **142** are configured to be attached to one another.
[0034] The segments may be identical or have different shapes. The shape of the segments is therefore adapted to the shape and the surrounding area of the optical surface **13**. The segments **14** form modules; the segments **14** are independent modules. In the device **10**, at least some of the segments are identical modules; this makes it possible to assemble the device simply but also to replace faulty segments simply. This makes it easier to manufacture the device. According to FIG. **1**, all the segments are identical modules.
[0035] The segments **14** have a shape which is elongate between two ends. The first segment **141** and the second segment **142** are configured to be attached to one another so as to form a circular arc. A circular arc is a portion of a curve that is delimited by two points of this curve; a circular arc is a portion of the circumference of a circle with a center and a radius. The circular arc defines an axial direction Z which passes through the center of the circular arc, a radial direction Y which is along a radius of the circular arc, and a tangential direction X which is tangential to the portion of a circle of the circular arc. The circular arc may extend in a plane, which is to say a two-dimensional space, but may also be shaped such that the circular arc is partially in a plane and extends in three

dimensions. The specific shape of each segment **14** depends on the shape of the optical surface **13**. At least some of the segments are in the form of a circular arc, such as the first segment **141** and the second segment **142**. This makes it possible to at least partially get round the optical surface **13**.

[0036] FIG. **2** shows an example of a segment **14**. The segment **14** has the shape of a circular arc, as defined above. As mentioned, the circular arc formed by the segments **141** and **142** may extend in three dimensions, and one segment **14** may have a slope between its ends. In a bottom view, the segment **14** is still a circular arc. The shape of the segment also depends on the area surrounding the optical surface **13** and is adapted to how crowded the optical surface **13** is. The segments **14** may have a circular-arc shape which differs from one segment to the next. This makes it possible to adapt it to the shape and the surrounding area of the optical surface **13**.

[0037] Some segments **14** may have a circular-arc shape and some segments may have a rectilinear shape or indeed any shape at all. The shape of the device is such that the segments face the optical surface **13**. The optical surface **13** is at least partially surrounded by the device **10** (at least in the portions by means of which the one or more sensors **12** interact with the surrounding area). The device **10** may have a closed shape (as in FIG. **1**) or an open shape. The device **10** may have a circular overall shape with the directions X, Y, Z that was defined above. According to FIG. **1**, it is possible to envisage that the segments **14** form an annular structure. The annular structure (of 360°) may have two segments attached to one another in pairs (each segment representing 180°), or three segments attached to one another in pairs (each segment representing 120°), or four segments attached to one another in pairs (each segment representing 90°). According to FIG. **1**, the segments **14** form a circle around the optical surface **13**, which has a cylindrical shape. The directions X and Y are in the same plane (that of the device **10**), with the tangential direction X extending tangentially to the circle and the radial direction Y extending radially toward the center of the circle; the axial direction Z is orthogonal to the directions X and Y, through the center of the circle. The direction Z may extend along the height of the optical surface **13**.

[0038] The segments **14** each have a cleaning fluid circulation channel **16**. The channel **16** in the segments is a flow duct with a hollow and elongate form, enabling passage of the fluid for cleaning the optical surface **13**. The channel **16** follows the shape of the segment **14**, which is to say a circular arc as described above. According to FIG. **2**, the first segment **141** shown comprises a first circulation channel **16** and the second segment **142** has a second circulation channel **16**. The same applies to any other additional segment **14**. Each channel **16** extends over at least some of the length of the respective segment **14**, but does not extend from one segment to the next; a channel **16** defines a flow duct specific to each segment **14**. Since the channels **16** are independent of one another, the device may have an open shape, such as an open loop.

[0039] In addition, the segments **14** each comprise a cleaning fluid inlet **18** and at least one nozzle **20** for diffusing the cleaning fluid toward the optical surface **13**. The inlet **18** is connected to a cleaning fluid distribution network and makes it possible to supply cleaning fluid to the channel **16**. Each segment **14** therefore comprises its own fluid inlet **18**; fluid is then supplied to each of the segments **14** independently. The inlet **18** may extend along an axis Z, but it is possible to envisage another shape, such as a curved shape, to adapt to the area surrounding the sensor. The channel **16** of each segment **14** then makes it possible to distribute cleaning fluid to the one or more fluid diffusion nozzles **20**. Five nozzles **20** are shown by way of example on the segment **14** in FIG. **2**; also by way of example, the segments **14** in FIG. **1** each have four nozzles **20**. The segments have a number of nozzles **20** which is specific to them, depending on the location of the segments **14** in relation to the optical surface **13** and in relation to the portion of optical surface **13** that is to be cleaned. Similarly, the nozzles **20** are distributed over each segment **14** depending on the portion of optical surface **13** that is to be cleaned.

[0040] The segments **14** also have a connector **22**. The first segment **141** has a first connector **22** and the second connector **142** has a second connector **22**. The same applies to any additional segment **14**. The segments **14** are configured to be attached in pairs, and depending on the segments **14**, to form a circular arc. The segments **14** are attached to one another by way of their respective connector **22**. The first connector **22** and the second connector **22** have complementary shapes and are configured to interact so as to make it possible to lock the first and second segments **141**, **142** relative to one another in the axial direction Z and in the radial direction Y. As a result, the segments **141** and **142** are attached to one another so as to form the device **10**. The interaction between the connectors **22** makes it possible to immobilize one segment relative to the other so as to make up the device **10**. The locking of the segments **14** makes it possible to give the device **10** a one-piece nature so as to be able to subsequently assemble it in position in the vehicle. The device therefore comprises modular segments each having a connector, the interaction between which provides the one-piece nature and ensures easy assembly around the optical surface **13**. In addition, the segments **14** are modular elements that can be replaced if a defect arises—notably when the segments are being assembled to one another. This makes it possible to correct the device **10** when it is being assembled without scrapping the device **10** entirely; all that is necessary is to replace the one or more defective segments.

[0041] The device **10** may have a number of segments that depends on the optical surface **13** to be cleaned. The segments **14** may then comprise two connectors **22**, one connector **22** at each end of the segments **14**. This makes it possible to join a segment **14** to two other, neighboring segments **14**. According to FIG. **1**, each segment **14** is joined to two neighboring segments **14** by its two connectors **22**. The device **10** can thus form a closed loop. However, the device **10** may be an open loop, the connector **22** of an end segment **14** not being joined to a neighboring segment.

[0042] The connectors **22** may make it possible for two neighboring segments **14** to be detachably engaged. This makes it possible to repair the device **10** after it has been assembled and while it is being used—without scrapping all of the device **10**. A faulty segment can be replaced. The connectors **22** may make it possible to snap-fasten two neighboring segments **14**. In other words, the connectors **22** form a clip. This enables easy assembly of the connectors **22** with one another and detachable locking of the segments.

[0043] FIG. **3** to FIG. **7** show examples of connectors **22**. The connectors **22** have complementary shapes for fixing one segment to the next.

[0044] FIG. **3** to FIG. **6** show the connectors **22** comprising a projection **26** on one of the segments and a bore **28** in another channel. The engagement of the projection **26** in the bore **28** prevents the segments **141**, **142** from moving in translation relative to one another along the radial direction Y and the tangential direction X. The first and second connectors **22** interact so as to make it possible to lock the segments **141**, **142** in relation to one another in a plane containing X and Y. The connectors **22** engage one in the other along the direction Z.

[0045] According to FIG. **3** and FIG. **4**, the projection **26** can be snap-fastened in the bore **28**, the snap-fastening of the projection **26** in the bore **28** preventing the segments **141**, **142** from moving in translation relative to one another along the axial direction Z. As a result, the first and second connectors **22** interact so as to make it possible to lock the segments **141**, **142** in relation to one another in the three directions X, Y, Z. The snap-fastening is obtained by shoulders **30** on the connector **22** of one of the segments **14** (the segment **142** in FIG. **3** and FIG. **4**) that engage with a reduction in diameter **32** on the other of the segments **14** (the segment **141** in FIG. **3** and FIG. **4**). The shoulders **30** and the reduction in diameter **32** form a clip, ensuring easy engagement. Once the shoulders **30** are engaged with the reduction in diameter **32**, the segments are secured. Furthermore, a slot **34** in one of the segments **14** (the segment **142** in FIG. **3** and FIG. **4**) makes the engagement of the connectors **22** detachable. A pressure exerted along the direction Y of the connector **22** against the slot **34** makes it possible to remove the connectors **22** from one another. This makes it possible to dismount the two segments **141**, **142** and to replace one of them if required.

[0046] According to FIG. **3** and FIG. **4**, the connectors **22** additionally comprise pads **36** and/or lugs **38** on at least one of the segments **14** that interact with the other segment and prevent the segments **14** from rotating relative to one another about the axial direction Z. This gives the device

10 a one-piece nature, avoiding deformation while the device is being mounted in the vehicle and throughout the service life of the device **10**.
[0047] More specifically, FIG. **3** presents pads **36** on the connector **22** of each of the segments **141**, **142**. The pads **36** on one segment **14** abut the end of the other segment **14**. The abutment of the end of one segment **14** against the pads **36** prevents the segments from rotating relative to one another about the direction Z. For example, the connectors **22** of each of the segments **141**, **142** have two pads **36** on a face that faces toward the other segment. The two pads **36** on one of the connectors **22** abut the end of the other segment **14** and prevent the segments from rotating relative to one another about the direction Z.
[0048] FIG. **4** presents lugs **38** on at least one of the connectors **22**. The lugs **38** on one segment **14** engage with the connector **22** of the other segment. The other segment may have slots **40** through which the lugs **38** engage. The engagement of the lugs **38** with the other segment prevents the segments from rotating relative to one another about the direction Z. For example, the connector **22** of the segment **141** has four lugs **38** around the bore **28** on a face that faces toward the other segment **142**. The four lugs **38** engage in the corresponding slots **40** of the other segment **142** and prevent the segments from rotating relative to one another about the direction Z.
[0049] According to FIG. **5** and FIG. **6**, the projection **26** (essentially hidden behind the connector **22** in FIG. **6**) can engage in the bore **28**, with the engagement of the projection **26** in the bore **28** preventing the segments **141**, **142** from moving in translation relative to one another along the directions X and Y. In addition, the interaction between the connectors **22** is elastic, and the connectors **22** may comprise tabs **42** on at least one of the segments **14** (the segment **141** in FIG. **5** and FIG. **6**) that engage with the other segment **14** (the segment **142** in FIG. **5** and FIG. **6**) and prevent the segments from moving in translation relative to one another along the axial direction Z. As a result, the first and second connectors **22** interact so as to make it possible to prevent the segments **141**, **142** from moving relative to one another in the three directions X, Y, Z. The tabs **42** form a clip, ensuring easy engagement. The tabs **42** also make the engagement of the connectors **22** detachable. Once the tabs **42** are engaged with the other segment, the segments are detachably secured. A spacing between the tabs **42** along the direction Y makes it possible to remove the connectors **22** from one another. This makes it possible to dismount the two segments **141**, **142** and to replace one of them if required. Furthermore, the other segment may have a notch **44** (the segment **142** in FIG. **5** and FIG. **6**) in which the tabs **42** can engage.
[0050] More specifically, FIG. **5** presents lugs **38** on at least one of the connectors **22**. The lugs **38** on one segment **14** engage with the other segment. The lugs **38** can engage in the notch **40** of the other segment. The engagement of the lugs **38** with the other segment prevents the segments from rotating relative to one another about the axial direction Z. For example, the connector **22** of the segment **141** has four lugs **38**, in pairs on either side of the tabs **42** on a face that faces toward the other segment **142**. The four lugs **38** engage in the corresponding notch **40** of the other segment **142** and prevent the segments from rotating relative to one another about the axial direction Z. The lugs **38** give the device **10** a one-piece nature, avoiding deformation while the device is being mounted in the vehicle and throughout the service life of the device **10**.
[0051] FIG. **6** presents a stud **46** on at least one of the segments **14** (for example the segment **141** in FIG. **6**) and an orifice **48** in the other segment **14** (for example the segment **141** in FIG. **6**). The engagement of the stud **46** in the orifice **48** and the snap-fastening of the projection **26** in the bore **28** prevent the segments from rotating relative to one another about the axial direction Z. This gives the device **10** a one-piece nature, avoiding deformation while the device is being mounted in the vehicle and throughout the service life of the device **10**.
[0052] According to FIG. **7**, the connector **22** of one of the segments **14** (for example, the segment **141** according to FIG. **7**) comprises a tongue **50**. The other segment **14** (for example, the segment **142** according to FIG. **7**) comprises a receiving portion **52**. The tongue **50** can engage in the receiving portion **52**, the snap-fastening of the tongue **50** in the receiving portion **52** preventing the

segments from moving in translation relative to one another along the tangential direction X, radial direction Y and axial direction Z. This gives the device **10** a one-piece nature, avoiding deformation while the device is being mounted in the vehicle and throughout the service life of the device **10**. As a result, the first and second connectors **22** interact so as to make it possible to lock the segments **141**, **142** in relation to one another along the tangential direction X, radial direction Y and axial direction Z and about the axial direction Z. The connectors **22** engage one in the other along the direction X, which is marked by the arrow **54**.
[0053] To ensure the two segments **141** and **142** are prevented from moving in translation relative to one another once the tongue **50** is engaged in the receiving portion **52**, the connector **22** provided with the tongue **50** may comprise clamps **56** with a slot, each interacting with a rib **58** of the other connector **22**. This makes it possible to detachably snap-fasten the connectors **22**. A pressure exerted along the direction Y of the clamps **56** against the slot makes it possible to remove the connectors **22** from one another. This makes it possible to dismount the two segments **141**, **142** and to replace one of them if required.
[0054] The segments **14** may comprise a passage **24** for an attachment member for attaching the cleaning device **10** to a sensor support **12**, the passage **24** going through the connectors **22**. The passage **24** through a respective connector **22** is an interface for attaching the device **10** to the support; it is not necessary to provide other attachment means, such as attachment tabs, and this simplifies the device **10**. In addition, the attachment member of the device **10** extending through the connectors **22** makes it possible to enhance the locking of the segments in relation to one another and avoids the risk of the segments **14** coming apart during use owing to vibrations in the vehicle. According to FIG. **3** to FIG. **6**, the passage **24** goes through the projection **26** and the bore **28**; once the projection **26** is engaged in the bore, the passage **24** of each connector **22** is coaxial with the passage **24** of the other connector. According to FIG. **7**, the passage **24** goes through the tongue **50** and the receiving portion **52**; once the tongue **50** is engaged in the bore **52**, the passage **24** of each connector **22** is coaxial with the passage **24** of the other connector. This makes it possible to insert an attachment member **10** of the device in the area surrounding the sensor **12**. The attachment member is for example a screw or a pad forming a clip.
[0055] The invention also relates to the detection system **11** shown in FIG. **1**, comprising the optical sensor **12** of a vehicle and the cleaning device **10**. The device **10** is configured to clean the optical surface **13** of the optical sensor. According to one embodiment, the optical surface **13** of the optical sensor **12** is cylindrical, the segments **14** forming an annular structure around at least one part of the circumference of the optical surface. The advantages described of the device **10** apply to the system **11**.
[0056] The present invention has been described in relation to specific embodiments, which have purely illustrative value and should not be considered limiting. In general, it will be obvious to a person skilled in the art that the present invention is not limited to the examples illustrated and/or described above.

Claims

1. A device for cleaning an optical surface of an optical sensor of a vehicle, the device comprising: a first segment having a first cleaning fluid circulation channel and a first connector, a second segment having a second cleaning fluid circulation channel and a second connector, the first and second segments each including cleaning fluid inlet and at least one nozzle for diffusing the cleaning fluid toward the optical surface, the first segment and the second segment being configured to be attached to one another so as to form a circular arc defining an axial direction passing through the center of the circular arc and a radial direction along a radius of the circular arc, the first connector and the second connector having complementary shapes and being configured to interact so as to make it possible to lock the first and second segments relative to one

another in the axial direction and in the radial direction.
2. The device as claimed in claim 1, wherein the connectors of segments attached to one another form a detachable snap-fastening means.
3. The device as claimed in claim 1, wherein the connectors include a projection on one of the segments and a bore in another segment, the engagement of the projection in the bore preventing the segments from moving in translation relative to one another in the radial direction and a tangential direction, which is orthogonal to the radial direction and axial direction.
4. The device as claimed in claim 3, wherein the projection can be snap-fastened in the bore, the snap-fastening of the projection in the bore preventing the segments from moving in translation relative to one another in the axial direction.
5. The device as claimed in claim 3, wherein the connectors additionally include pads and/or lugs on at least one of the segments that interact with the other segment and prevent the segments from rotating relative to one another about the axial direction.
6. The device as claimed in claim 3, wherein the connectors include tabs on at least one of the segments that engage with the other segment and prevent the segments from moving in translation relative to one another in the axial direction.
7. The device as claimed in claim 6, wherein the connectors additionally comprise lugs on at least one of the segments that engage with the other segment and prevent the segments from rotating relative to one another about the axial direction, and/or a stud on at least one of the segments and an orifice in the other segment, the engagement of the stud in the orifice and the snap-fastening of the projection in the bore preventing the segments from rotating relative to one another about the axial direction.
8. The device as claimed in claim 1, wherein the connector of one of the segments includes a tongue which engages in a receiving portion of the other connector of the other segment, the snap-fastening of the tongue in the receiving portion preventing the segments from moving in translation relative to one another in the tangential direction, radial direction and axial direction and about the axial direction.
9. The device as claimed in claim 1, further comprising a passage for an attachment member for attaching the cleaning device to a sensor support, the passage going through the connectors.
10. The device as claimed in claim 1, wherein the segments form an annular structure, the structure having two segments attached to one another in pairs.
11. The device as claimed in claim 1, wherein at least some of the segments are identical modules.
12. A detection system comprising an optical sensor of a vehicle and a cleaning device, the cleaning, device being configured to clean the optical surface of the optical sensor and including a first segment having a first cleaning fluid circulation channel and a first connector, a second segment having a second cleaning fluid circulation channel and a second connector, the first and second segments each including a cleaning fluid inlet and at least one nozzle for diffusing the cleaning fluid toward the optical surface, the first segment and the second segment being configured to be attached to one another so as to form a circular arc defining an axial direction passing through the center of the circular arc and a radial direction along a radius of the circular arc, the first connector and the second connector having complementary shapes and being configured to interact so as to make it possible to lock the first and second segments relative to one another in the axial direction and in the radial direction.
13. The system as claimed in claim 12, wherein the optical sensor has a cylindrical optical surface the segments forming an annular structure around at least one part of the circumference of the optical surface.
14. The device as claimed in claim 1, wherein the segments form an annular structure, the structure having three segments attached to one another in pairs.
15. The device as claimed in claim 1, wherein the segments form an annular structure, the structure having four segments attached to one another in pairs.